WHERE TO **BUILD**, WHERE TO **PRESERVE**

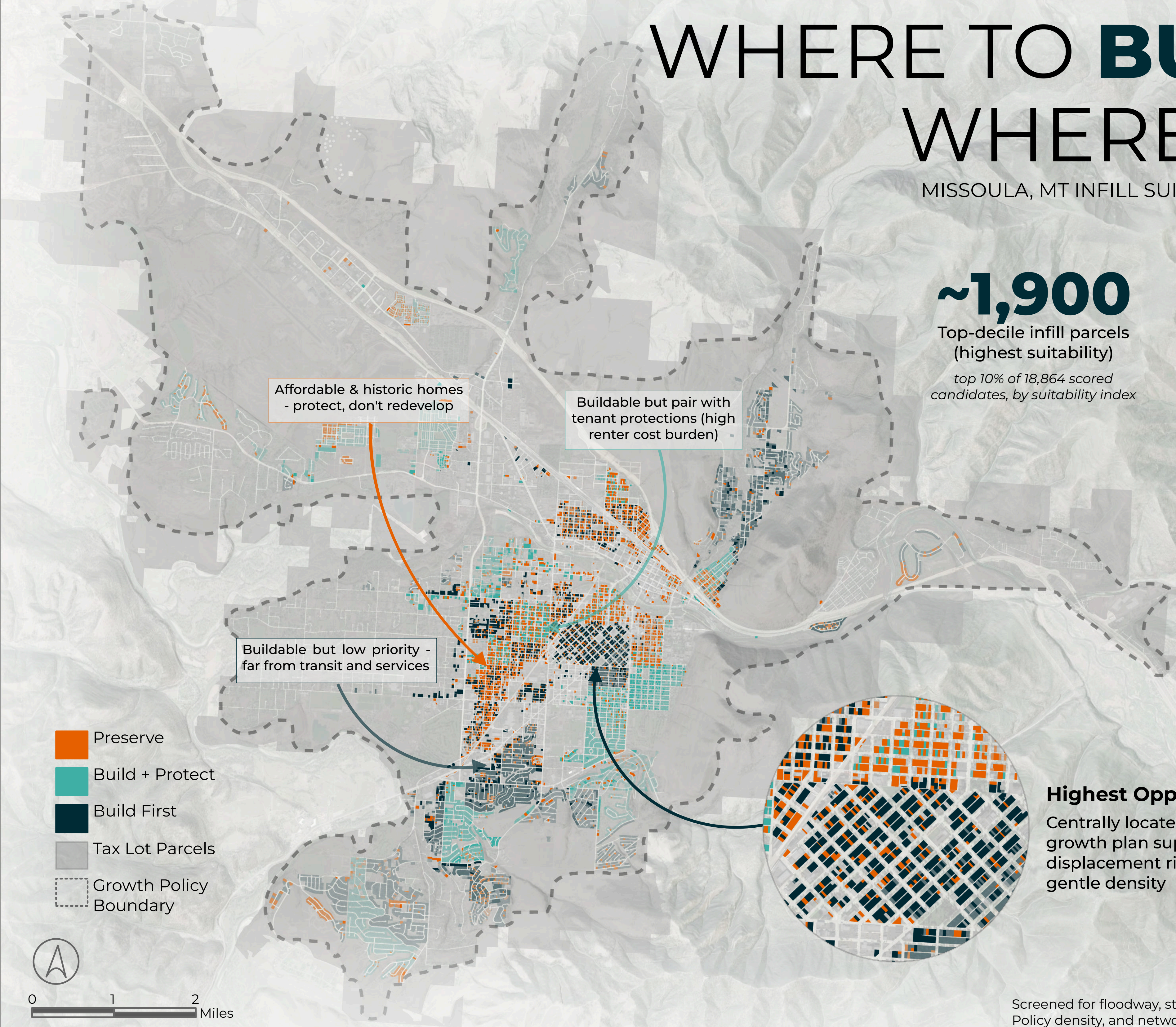
MISSOULA, MT INFILL SUITABILITY   2024 TAXLOT ANALYSIS   JASON FAN

~1,900
Top-decile infill parcels (highest suitability)
top 10% of 18,864 scored candidates, by suitability index

62%
Flagged for preservation, not redevelopment
historic, mobile-home, or naturally-affordable (NOAH) stock

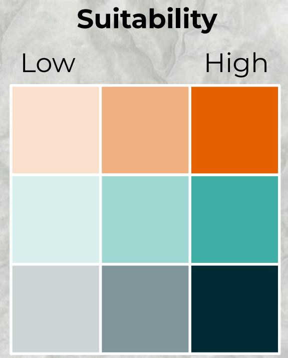
717
Net build-ready parcels (Build First & Build + Protect)
471 Build First parcels & 246 Build + Protect parcels

Missoula's best infill land is already doing affordability work, so the move is targeted growth and preservation, not a blanket upzone.



Highest Opportunity Cluster

Centrally located, transit served, growth plan supported, low displacement risk. Ready for gentle density



Preserve:
Keep existing affordable or historic homes

Build + Protect:
Add homes, safeguard renters with tenant protections

Build First:
ready for gentle density. Low displacement risk

Suitability (light→dark) shows where infill fits; color shows what to do there

Screened for floodway, steep slope, and 1,255 contributing/listed historic structures. Scored on under-utilization, capacity, Growth-Policy density, and network walk-access. Source: Missoula 2024 taxlots · City historic preservation · Census ACS · OpenStreetMap.

Part 2: Strategic Project Management

2a. Scaling & Technical Workflow

Primary hurdle: **cross-jurisdiction data normalization**.

The MSDI cadastral gives us parcel geometry and the ORION assessment schema countywide, but not the fields the model needs. Zoning, future land use, and servicing are held separately by the City and County in systems that do not align, and buildability outside the City often depends on septic feasibility under DEQ review rather than sewer. Before any analysis, these sources have to be harmonized into one regional parcel layer.

My approach is a documented crosswalk feeding an automated pipeline.

1. Inventory every source layer and record its schema, coverage, and licensing.
2. Define one target schema and a crosswalk that maps each jurisdiction's codes into it. This is the highest-risk step, so I own it.
3. Build a scripted, version-controlled pipeline that applies the crosswalk and outputs the harmonized layer, so monthly refreshes re-run without rework.
4. Re-point the Missoula scoring model at the regional layer and recalibrate factors that do not travel, such as transit access.
5. Publish the scored layer as a dashboard with per-jurisdiction filters and a documented methodology.

Steps 1 and 2 are the critical path and gate everything else, so I front-load and staff them to protect the schedule.

2b. Delegation & Quality Control

I stay on the strategy and the calls that carry defensibility risk, and push production work down to keep senior hours low. I handle scoping, crosswalk design, model decisions, client communication, and final sign-off. The Junior Associate builds and documents the pipeline, ports the model, builds the dashboard, writes the validation tests, and owns reproducibility. The GIS Intern does data collection, first-pass attribute mapping, geometry cleanup, and QA sampling, and comes to the Junior first for technical questions. We track against a bi-weekly check-in and a short written status report.

QC runs as three gates, and nothing reaches the client until each clears and I sign off. Ingestion is reconciled against the source cadastral for counts, values, coordinate system, and topology. The crosswalk is validated on a blind sample of about 30 parcels per jurisdiction and has to hit 95% agreement before the model runs, with disagreements settled by me. The regional output is checked against the Missoula pilot as ground truth for the City slice and spot-checked on high-impact parcels.

2c. Budget & Schedule Management

Task / phase	Weeks	Sr Associate	Jr Associate	Intern	Cost
1. Data acquisition & inventory	1–2	10	24	34	$5,020
2. Regional data model & crosswalk	2–4	26	40	30	$8,700
3. Normalization pipeline (ETL)	4–6	12	78	46	$10,660
4. Model port + dashboard build	6–9	16	78	40	$11,020
5. QC & validation	8–10	12	30	40	$6,100
6. PM, documentation & delivery	10–11	24	10	0	$4,500
Labor subtotal (550 hrs)		**100**	**260**	**190**	**$46,000**
Non-labor: hosting, geocoding/QA credits, contingency					$4,000
Total					**$50,000**

The engagement is time-and-materials with a not-to-exceed fee of $50,000 over about 11 weeks. Each task has a deliverable tied to a milestone so the City can track progress against payment.

Blended rates (Assumed): Senior Associate $150/hr, Junior Associate $90/hr, GIS Intern $40/hr, part-time across the engagement. Phases overlap and QC runs alongside the build.

I hold the fixed fee two ways. Senior hours stay low because I keep production with the Junior and Intern and reserve my own time for design and review, and the non-labor line carries a contingency that I draw scope changes against in writing rather than absorbing them. The main schedule risk is late data handoffs from the City and County, so I start acquisition in week 1 and keep the critical-path steps, inventory and crosswalk, fully staffed, since everything downstream waits on them.

Sources: Montana State Library MSDI Cadastral framework, MT DEQ Sanitation in Subdivisions Act, and the Missoula Organization of REALTORS Five Valleys Housing Report.